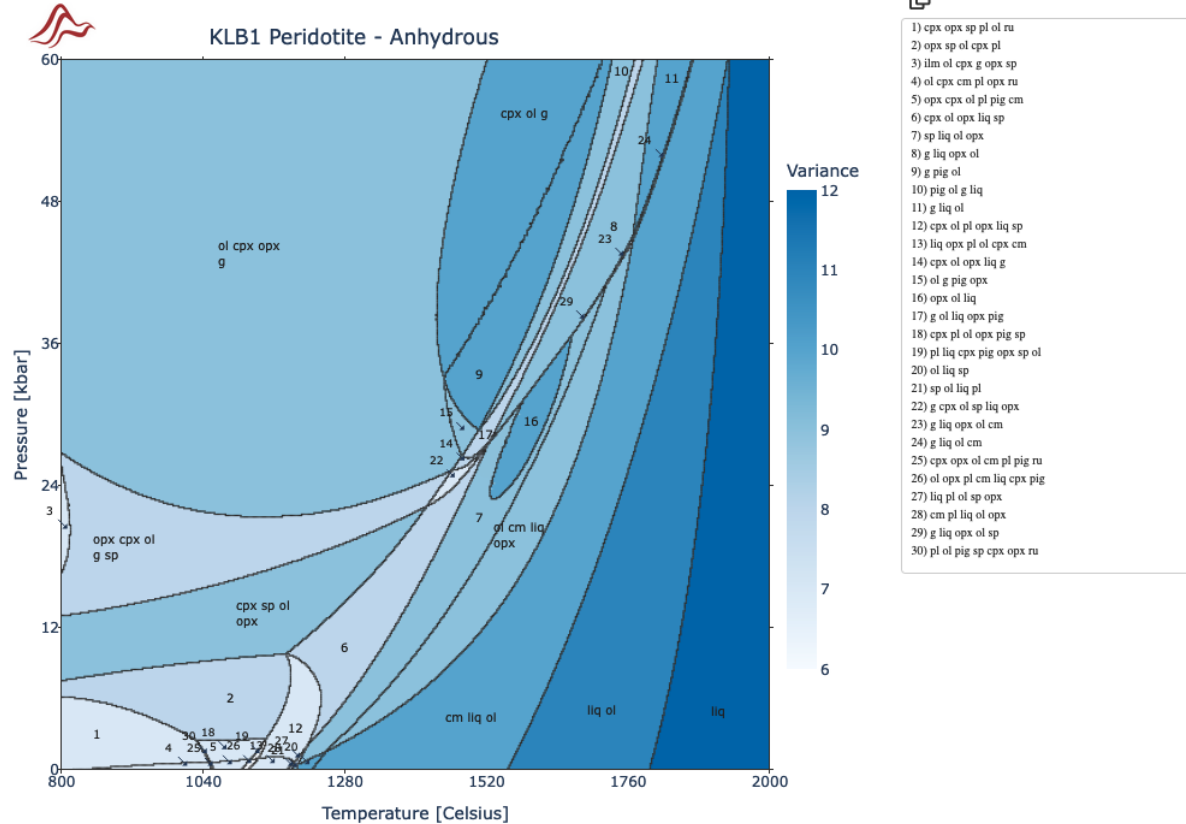


MAGEMinApp: phase diagrams tutorials



MAGEMinApp: phase diagrams interface

Phase diagrams

PTX path

Isentropic path

General information

Setup

Diagram

Trace-elements

Phase diagram parameters

Thermodynamic database

Igneous (Green et al., 2025, after H18)

Solution phase selection

Diagram type

P-T diagram

H₂O-saturated at solidus

false

Limit Ca-opx

0.5

TE predictive model

false

Pressure [kbar]

0.01

20.01

Temperature [°C]

800

1400

Initial grid subdivision

3

8 × 8 grid

Refinement type

Phases only

Refinement levels

2

Buffer

no buffer

Solver

Hybrid

Verbose

none

Specific Cp

G0

Bulk-rock composition

mol%

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide

fraction

SiO₂

Al₂O₃

CaO

MgO

FeO

K₂O

Na₂O

TiO₂

O

Cr₂O₃

0.109

H₂O

0

ig

General parameters

Title

KLB1 Peridotite - Anhydrous

Update

Reset

Compute phase diagram

Save/Load Diagram

filename

Save state

Load state

Phase diagrams

PTX path

Isentropic path

General information

Setup

Diagram

Trace-elements

Phase diagram parameters

Thermodynamic database

Igneous (Green et al., 2025, after H18)

Solution phase selection

MAGeMinApp: first P-T phase diagram

- Select the following setup (nearly default)

Phase diagrams | **PTX path** | Isentropic path | General information

Setup | **Diagram** | Trace-elements

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18) ▼

Solution phase selection: ☐

Diagram type: P-T diagram ▼

H₂O-saturated at solidus: false ▼

☐ Limit Ca-opx: 0.5 ▼

TE predictive model: true ▼

Kd's database: O. Laurent (2012) ▼

Zr saturation: Crisp and Berry (2022) ▼

Pressure [kbar]: min 0.01 ▼ max 20.01 ▼

Temperature [°C]: min 800 ▼ max 1400 ▼

Initial grid subdivision: 3 ▼ 8 × 8 grid

Refinement type: Phases only ▼

Refinement levels: 2 ▼

Buffer: no buffer ▼

Solver: Hybrid ▼

Verbose: none ▼

Specific Cp: G0 ▼

Bulk-rock composition

mol% ▼

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous ▼

oxide	fraction
SiO ₂	38.494
Al ₂ O ₃	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K ₂ O	0.01
Na ₂ O	0.25
TiO ₂	0.1
O	0.096
Cr ₂ O ₃	0.109
H ₂ O	0

ig

Trace-element composition

Drag and drop or select trace-element file

Pyrolite (Primitive Mantle) ▼

Trace-element composition

Drag and drop or select trace-element file

Pyrolite (Primitive Mantle) ▼

elements	μg/g
Rb	0.6
Ba	6.6

- Select Pyrolite for trace-element composition

General parameters

Title: KLB1 Peridotite - Anhydrous

Update Reset

Compute phase diagram

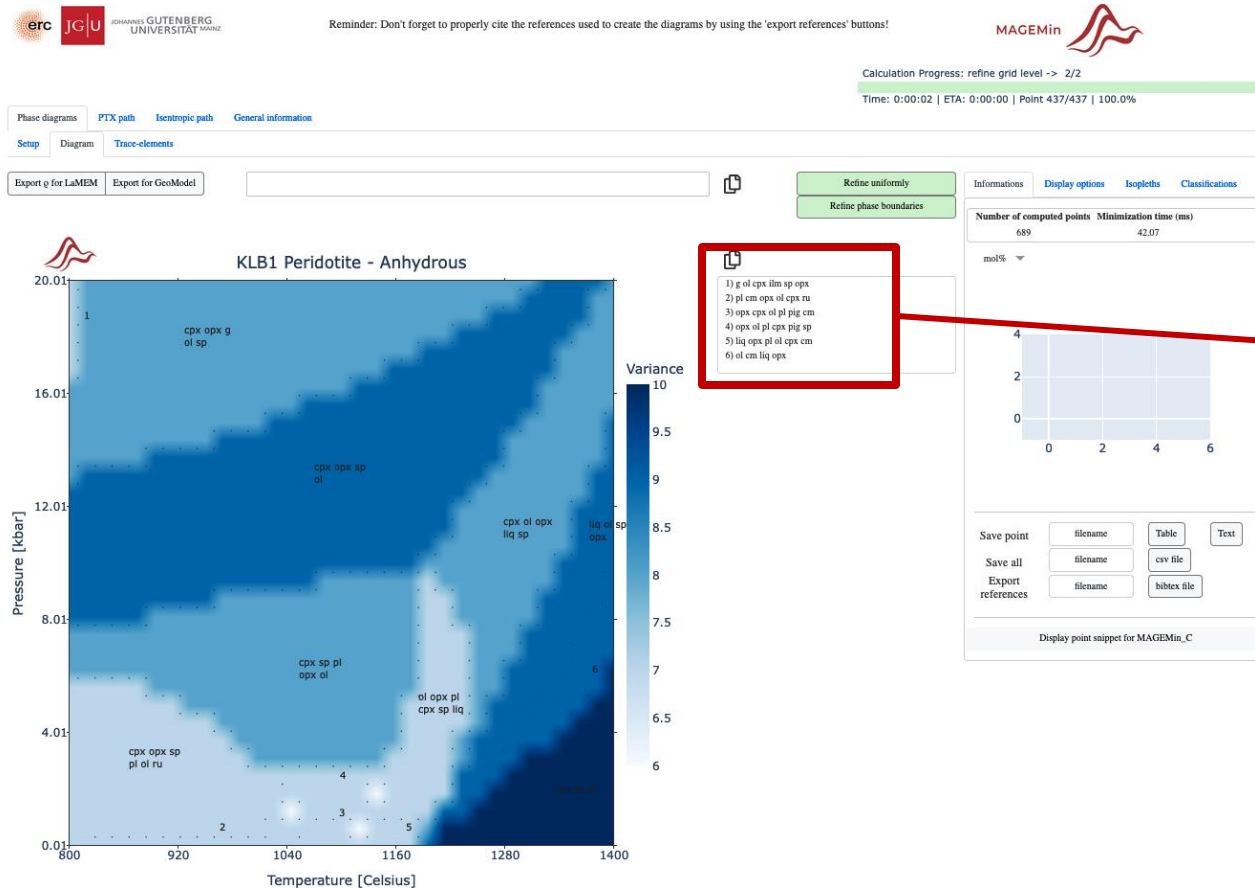
Save/Load Diagram

filename Save state Load state

- Click compute phase diagram

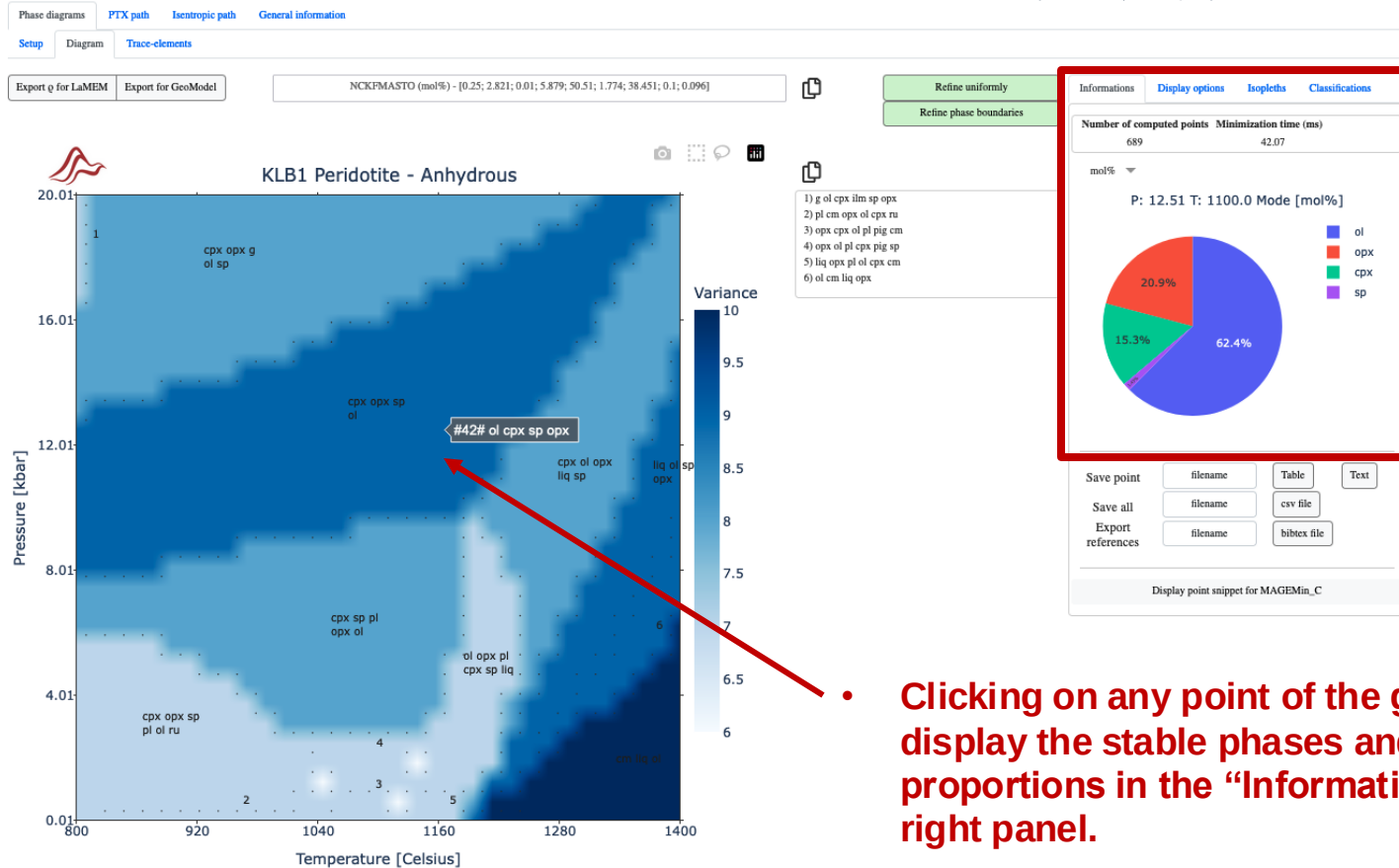
MAGeMinApp: first phase diagram

- Once computed, the low-resolution diagram should look like this (the tab is changed to diagram automatically):

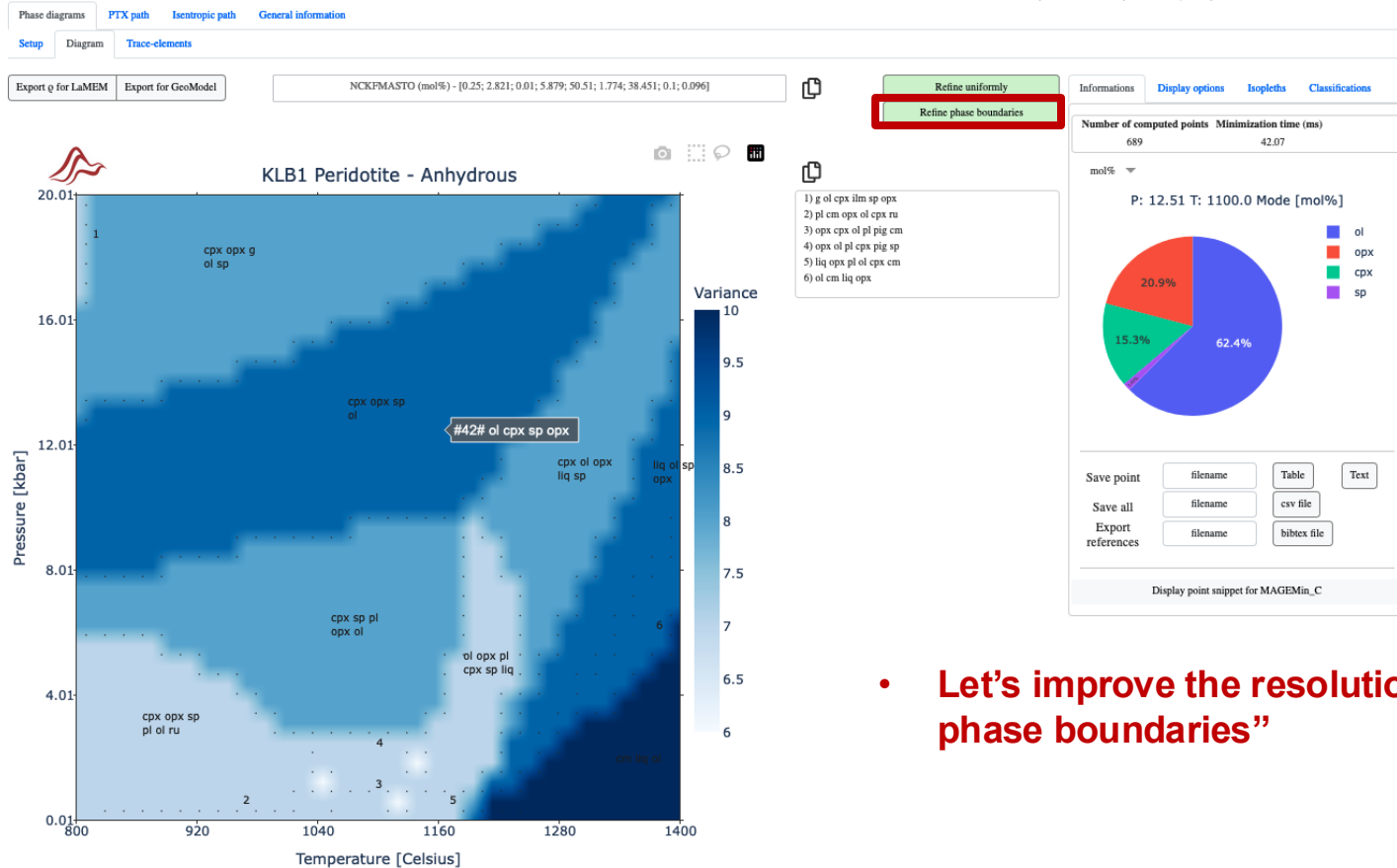


- List of stable phase mineral assemblage that cannot fit the size of the field

MAGeMinApp: first phase diagram

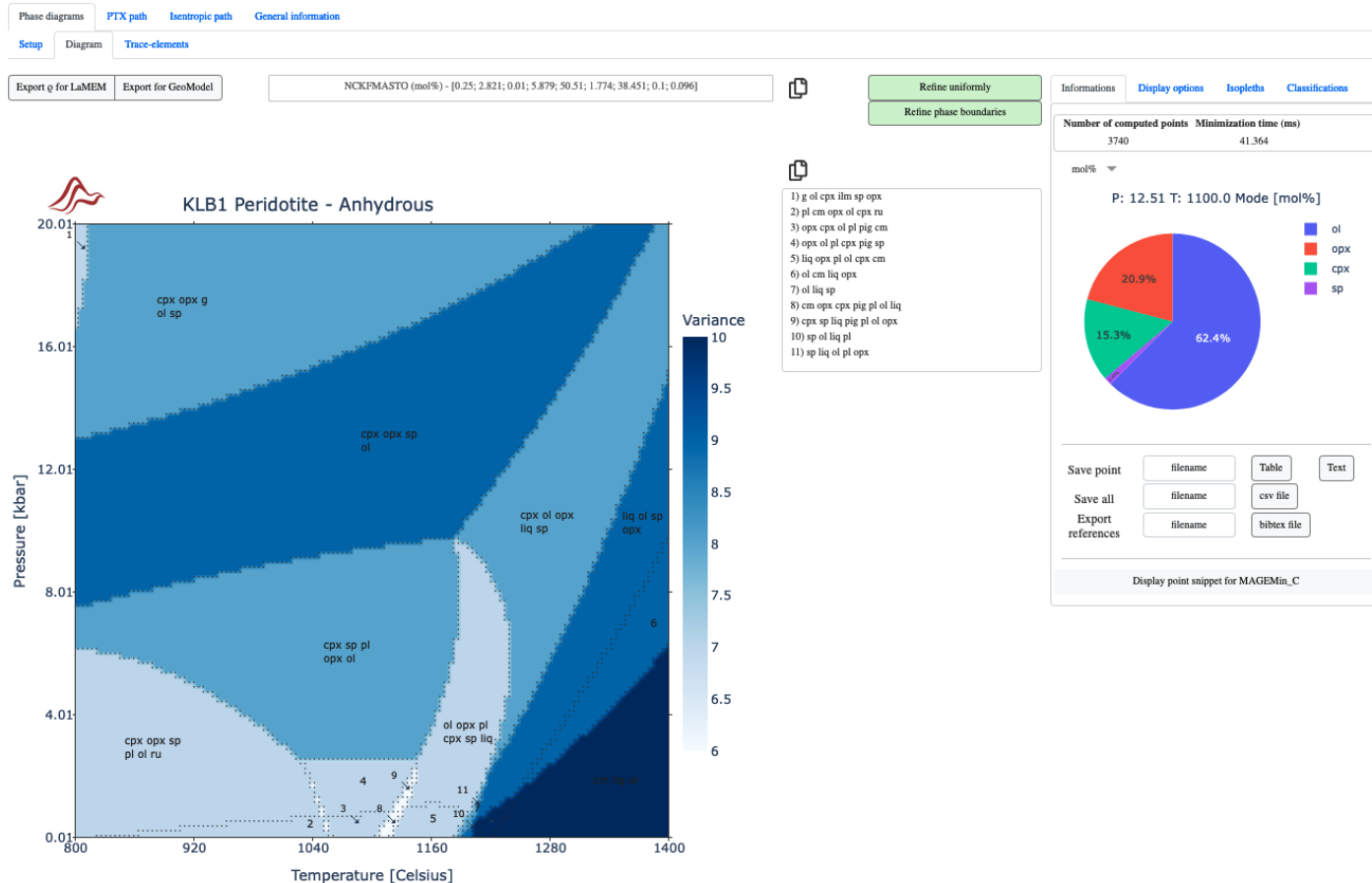


MAGeMinApp: first phase diagram



MAGeMinApp: first phase diagram

- After refining twice, the phase diagram starts to look better



MAGEMinApp: colormap range

- The bottom right panel allow you to change the display options

Informations Display options Isopleths Classifications

Select field to display

Field Variance ▼

Diagram options

Show reaction lines true ▼

Show stable phases true ▼

Show grid false ▼

Color options

Colormap Blues ▼

min max

Value range 6 10

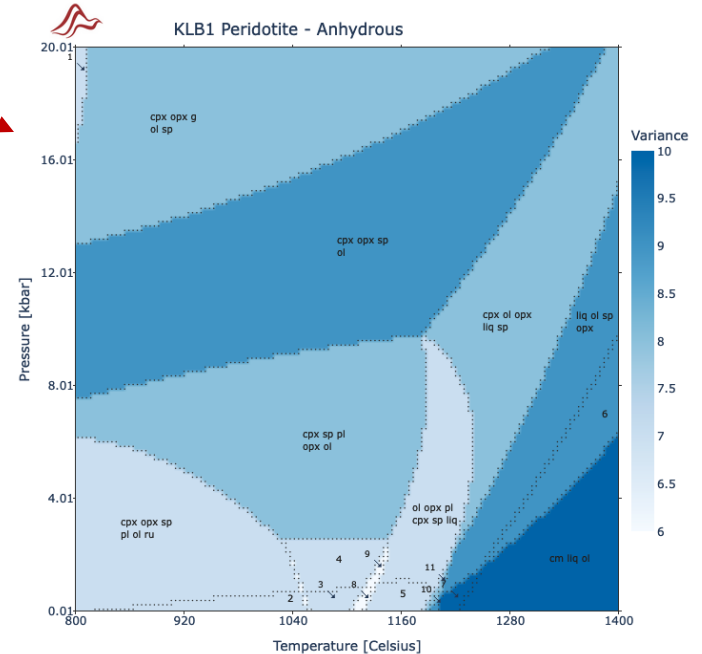
Colormap range 1 2 3 4 5 6 7 8 9

Set min to white false ▼

Reverse colormap false ▼

Smooth colormap best ▼

- Decreasing the colormap range allow the better visualize the high variance fields



MAGEMinApp: colormap

- You can of course change the colormap

Informations Display options Isopleths Classifications

Select field to display

Field: Variance

Diagram options

Show reaction lines: true

Show stable phases: true

Show grid: false

Color options

Colormap: Blues

Value range: min 6 max 10

Colormap range: 1 2 3 4 5 6 7 8 9

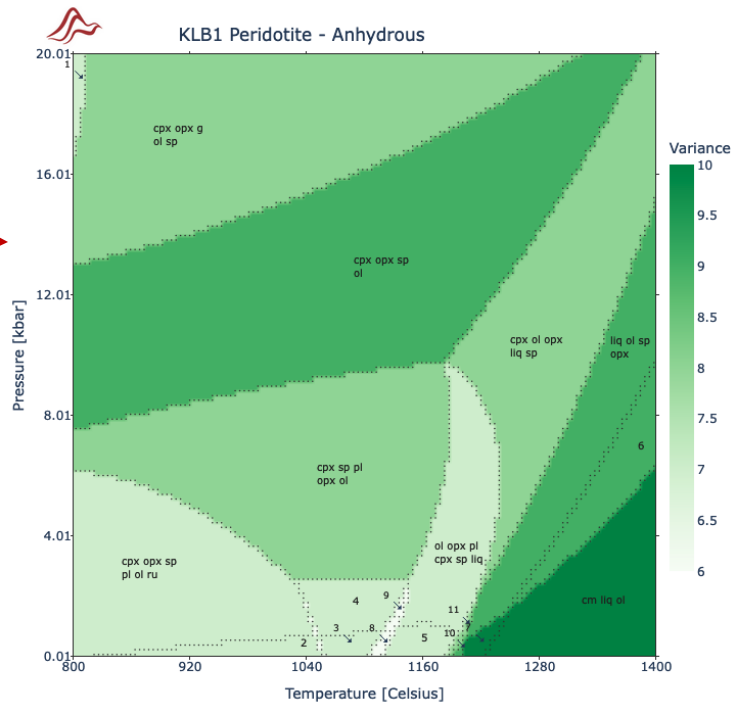
Set min to white: false

Reverse colormap: false

Smooth colormap: best

Colormap

Greens



MAGEMinApp: grid display

- display the grid...

Informations Display options Isopleths Classifications

Select field to display

Field Variance

Diagram options

Show reaction lines true

Show stable phases true

Show grid false

Color options

Colormap Blues

min max

Value range 6 10

Colormap range 1 2 3 4 5 6 7 8 9

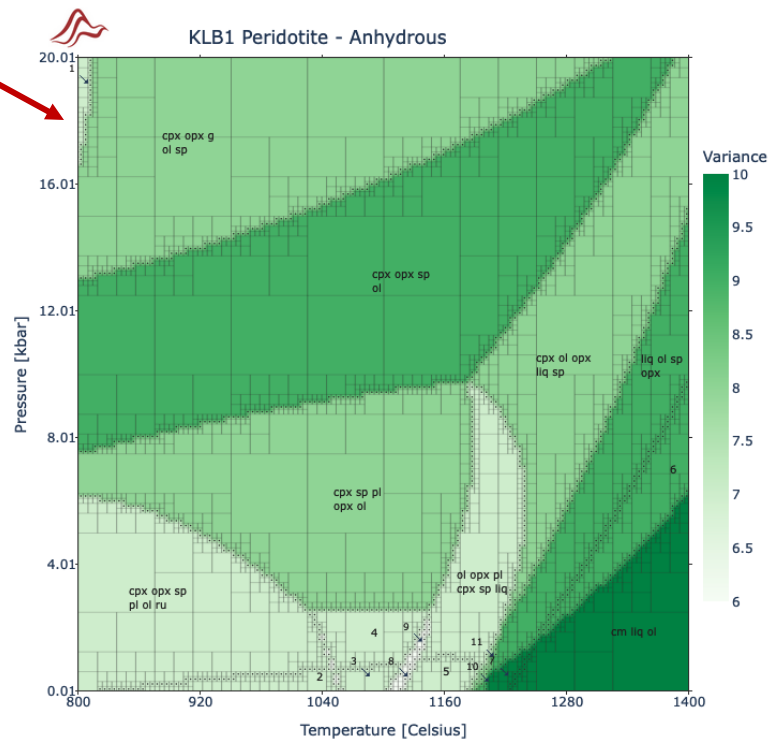
Set min to white false

Reverse colormap false

Smooth colormap best

Show grid

true



MAGMinApp: field

- change the field to be displayed

Informations Display options Isopleths Classifications

Select field to display

Field Variance

Diagram options

Show reaction lines true

Show stable phases true

Show grid false

Color options

Colormap Blues

min max

Value range 6 10

Colormap range 1 2 3 4 5 6 7 8 9

Set min to white false

Reverse colormap false

Smooth colormap best

Informations Display options Isopleths Classifications

Select field to display

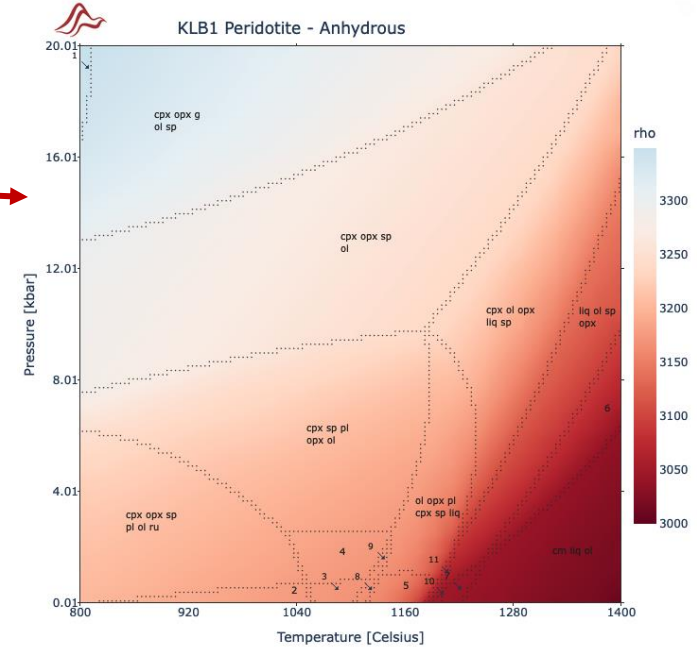
Field p_{system}

Diagram options

Show reaction lines true

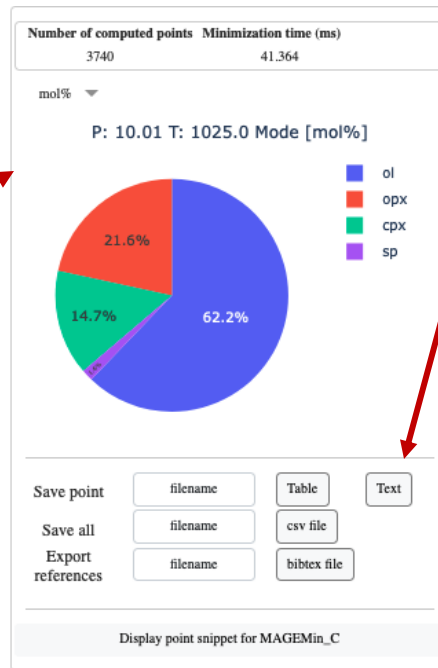
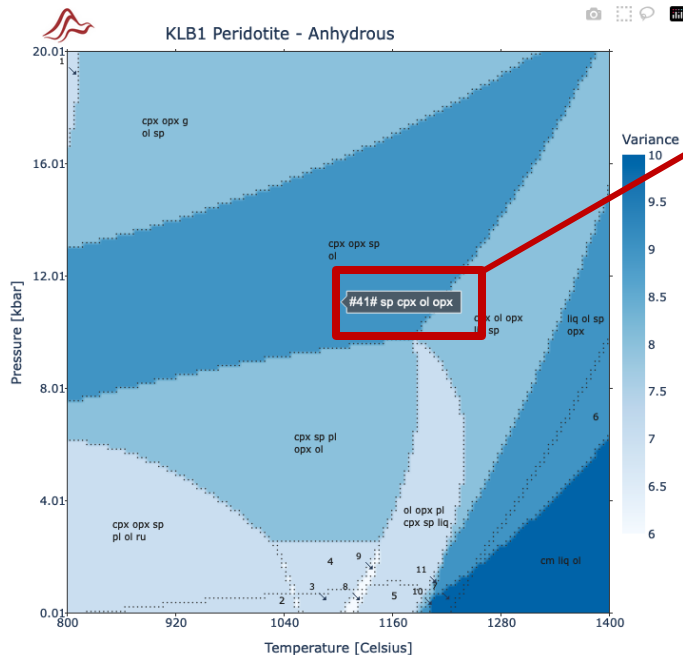
Show stable phases true

Show grid false



MAGEMinApp: saving data

- You can save point wise information or the whole diagram data using the right panel
- First, let's click on any grid point

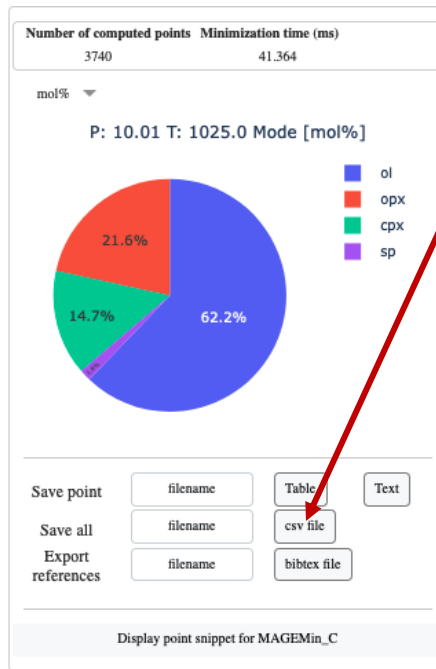
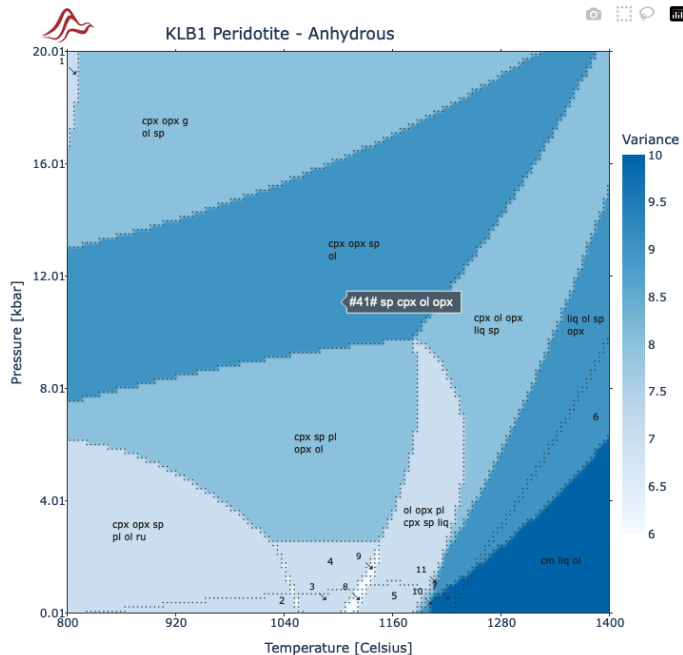


- Change the filename to “test” and click “text”
this will save a file in your download directory
e.g.,
test_ig_Pkbar_11.885000000000002_TC_1081.25.txt

```
=====
1  cpx.....ol.....spn.....opx.....{11.8850-1081.2500}..kbar/°C
2
3
4  End-members.fractions[wt.fr]:
5  .....di.....cfs.....cats.....crdi.....cess.....cbuf
6  0.596337 0.066411 0.012233 0.017320 0.022295 0.032956
7  .....mont.....fa.....fo.....cfm
8  0.003522 0.177025 0.819402 0.000051
9  .....nsp.....isp.....nhc.....ihc.....nmt.....imt
10 0.553107 0.231982 0.141464 0.021306 -0.013429 0.02323
11 .....en.....fs.....fm.....odi.....mgts.....crer
12 0.639219 0.053216 0.082200 0.045373 0.118856 0.023216
13
14 Oxide.compositions.[wt.fr]:
15 .....SiO2.....Al2O3.....CaO.....MgO.....FeO
16 SYS 0.448600 0.035124 0.030719 0.395275 0.082032
17 .....cpx 0.531572 0.054166 0.175962 0.168373 0.038439
18 .....ol 0.408187 0.000000 0.000952 0.490787 0.100073
19 .....spn 0.000000 0.636768 0.000000 0.230242 0.083187
20 .....opx 0.527534 0.074349 0.012709 0.309750 0.064255
21
22 Stable mineral.assemblage:
23 phase.....fraction[wt].....G[kJ].....V_molar[cm3/mol]
24 cpx.....+0.14461.....-3457.26908.....+66.64026
25 ol.....+0.62836.....-2339.30393.....+45.23134
26 spn.....+0.01071.....-2446.84718.....+41.03734
27 opx.....+0.21632.....-3373.92842.....+64.14469
```

MAGEMinApp: saving data

- You can save point wise information or the whole diagram data using the right panel
- First, let's click on any grid point



- Change the filename to “test” and click “csv file”
this will save all data to an csv file
- Check Julia terminal. Here the file is saved in a different directory (not through the browser) because it can become very big!

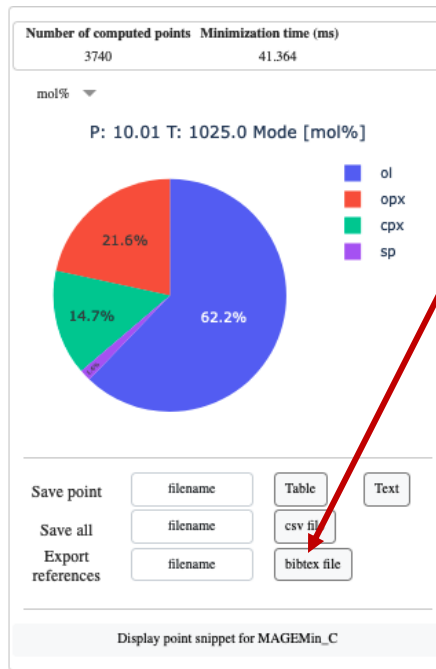
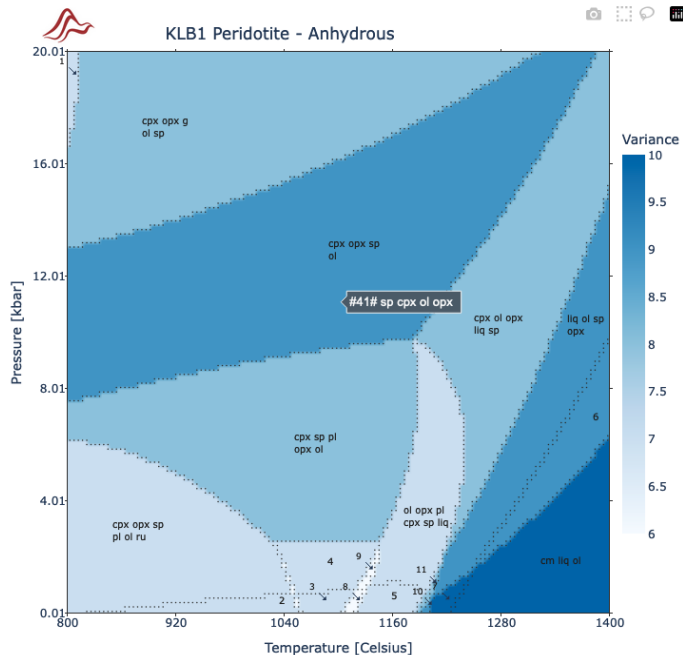
```
output path: /home/seph42/MAGEMinApp.jl_v04.0
Saving data to csv... 100%
```

- The “csv” file is a complete save of all information of the phase diagram. When the number of points becomes larges (e.g., > 50k points) it can take a few minutes to save a file bigger than a few Go.

	A	B	C	D	E	F	G	H	I	J	
1	point[#]	X[0.0-1.0]	P[kbar]	T[°C]	phase	model[mol]	model[wt%]	log10(fO2)	log10(dQ/aH2O)		a
2	1	1	0.635	818.75	system	100	100	-11.6936	2.559878	0	(
3	1	1	0.635	818.75	fsp	6.959184	6.970663	-	-	-	-
4	1	1	0.635	818.75	spn	0.174666	0.248471	-	-	-	-
5	1	1	0.635	818.75	cpx	6.119897	6.341448	-	-	-	-
6	1	1	0.635	818.75	ol	69.78836	69.61555	-	-	-	-
7	1	1	0.635	818.75	ol	16.87659	16.72111	-	-	-	-
8	1	1	0.635	818.75	ru	0.081307	0.102756	-	-	-	-
9	2	1	0.3225	846.875	system	100	100	-11.1585	2.557661	0	(
10	2	1	0.3225	846.875	fsp	6.946325	6.956501	-	-	-	-
11	2	1	0.3225	846.875	ol	69.78973	69.61939	-	-	-	-
12	2	1	0.3225	846.875	opx	16.94176	16.79384	-	-	-	-
13	2	1	0.3225	846.875	cpx	6.082929	6.30309	-	-	-	-
14	2	1	0.3225	846.875	spn	0.159418	0.226294	-	-	-	-
15	2	1	0.3225	846.875	ru	0.079834	0.100888	-	-	-	-
16	3	1	0.16625	860.9375	system	100	100	-10.9044	2.55316	0	(
17	3	1	0.16625	860.9375	ol	69.7901	69.62	-	-	-	-
18	3	1	0.16625	860.9375	fsp	6.938948	6.94901	-	-	-	-
19	3	1	0.16625	860.9375	opx	16.97846	16.83468	-	-	-	-
20	3	1	0.16625	860.9375	cpx	6.061894	6.281553	-	-	-	-

MAGEMinApp: saving data

- You can save point wise information or the whole diagram data using the right panel
- First, let's click on any grid point



You can also save the references used to perform the calculation clicking on “bibtex file”!

```

bib
article{holland2018melting,
pages
= {881–900},
info
= {ig},
author
= {Holland, Tim JB and Green, Eleanor CR and Powell, Roger},
journal
= {Journal of Petrology},
title
= {Melting of peridotites through to granites: a simple thermodynamic model in the system KNCFMASHOTC},
publisher
= {Oxford University Press},
number
= {15},
doi
= {10.1093/petrology/egy048},
year
= {2018},
volume
= {59}

article{riel2022magmin,
pages
= {2022GC0010427},
info
= {MAGEMin},
author
= {Riel, Nicolas and Kaus, Boris JP and Green, ECR and Berlie, Nicolas},
journal
= {Geochemistry, Geophysics, Geosystems},
title
= {MAGEMin, an efficient Gibbs energy minimizer: application to igneous systems},
publisher
= {Wiley Online Library},
number
= {7},
doi
= {https://doi.org/10.1029/2022GC0010427},
year
= {2022},
volume
= {23}
}

```

MAGMinApp: fields isocontours and isopleths

Informations Display options **Isopleths** ~~Classifications~~

Isopleth type: Pure phase ▼

Phase: ru ▼

Unit: mol ▼

Range

Min: 0 ▼

Step: 0.1 ▼

Max: 1 ▼

Plotting options

Line style: Solid ▼

Line width: 1 ▼

Color:

Label size: 10 ▼

Add

Displayed: Select... ▼ Hidden: Select... ▼

Hide Show

Hide all Show all

Remove

Remove all

• Click on the “Isopleths” panel

Pure phase ▲

Pure phase

Solution phase

Other

• Here you can select what to contour:
Pure phase, Solution phase or Other (fields)

• Here you can select min, step and max values

• Here you can select the linestyle, color and fontsize of the isocontour

• Here you add the isocontour

• And here you can manage the isocontour list

MAGMinApp: fields isocontours and isopleths

- Let's contour melt fraction

Informations Display options Isopleths Classifications

Isopleth type Solution phase

Phase liq

Field Mode

Unit mol

Range

Min 0

Step 0.1

Max 1

Plotting options

Line style Solid

Line width 1

Color

Label size 10

Add

Displayed Hidden

Select... Select...

Hide Show

Hide all Show all

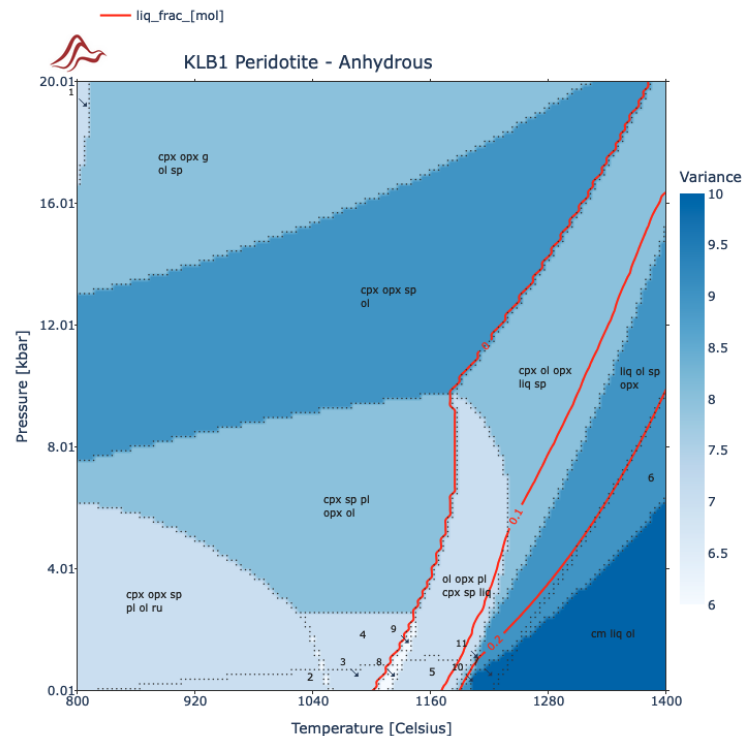
Remove

Remove all

- Select:
Isopleth type = "Solution phase"
Phase = "liq"
Endmember = "none"

- This will display the fraction in mol%

- Change the color, linestyle and fontstyle to what you like then click "add"



MAGMinApp: fields isocontours and isopleths

- Adding density contour

Informations Display options Isoleths Classifications

Isoleth type: Other

Field: ρ_{solid}

Range

Min: 3100

Step: 20

Max: 3300

Plotting options

Line style: Dot

Line width: 2

Color: 

Label size: 10

Add

Displayed: liq_frac_[mol]

Hidden: Select...

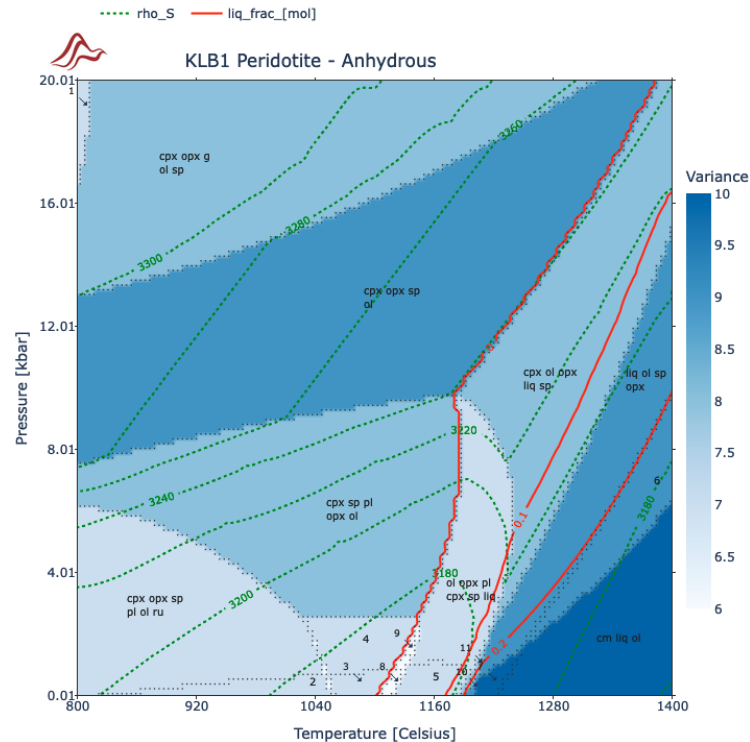
Hide Show

Hide all Show all

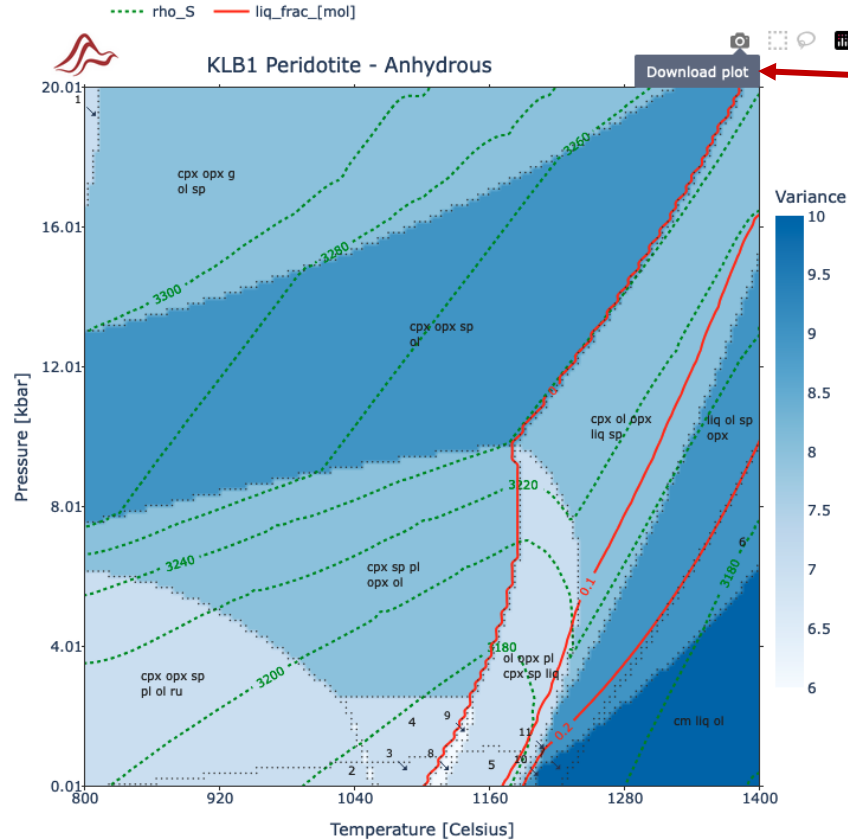
Remove

Remove all

- Select:
Isoleth type = "Other"
Phase = "rho_solid"
- Change range to 3100, 20, 3300
- Change the color to what you like then click add
- The "Displayed" and "Hidden" lists allow you to manage the isopleths

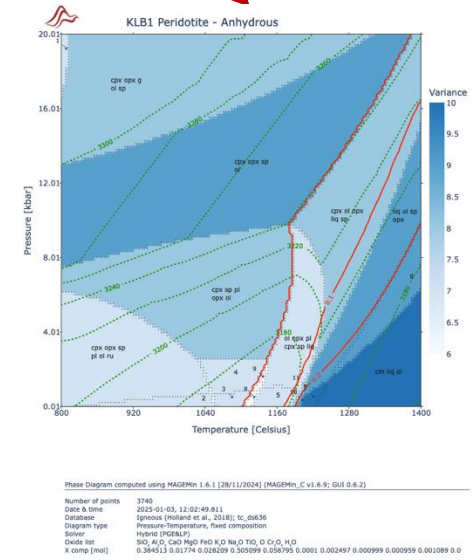


MAGMinApp: save figure(s)



- 1) g ol cpx ilm sp opx
- 2) ol cm opx ol cpx ru
- 3) opx cpx ol pl pig cm
- 4) opx ol pl cpx pig sp
- 5) liq opx pl ol cpx cm
- 6) ol cm liq opx
- 7) ol liq sp
- 8) cm opx cpx pig pl ol liq
- 9) cpx sp liq pig pl ol opx
- 10) sp ol liq pl
- 11) sp liq ol pl opx

Click here to save the plot as an “svg” vector file that can be opened in Inksape



- All plots and figures in the app can be saved in the same way

MAGEMinApp: export to LaMEM

- Let's first compute a P-T diagram for a metasediment

Phase diagram parameters

Thermodynamic database	Metapelite (White et al., 2014)
Solution phase selection	☐
Diagram type	P-T diagram
H ₂ O-saturated at solidus	false
TE predictive model	false
Pressure [kbar]	min: 0.01, max: 30
Temperature [°C]	min: 100, max: 1400
Initial grid subdivision	6, 64 x 64 grid
Refinement type	Phases only
Refinement levels	0
Buffer	no buffer
Solver	Hybrid
Verbose	none
Specific Cp	G0

id=5dd605fb-19e9-4933-ad9f-0c828d2532fa; MAGEMinApp GUI v=0.6.2; using 12/12 threads

- Select:
Database = Metapelite
Pressure = 0.01 to 30.0 kbar
Temperature = 100 to 1400 °C
Initial Grid Sub. = 6
Refinement levels = 0

- Choose FPWorldMedian pelite – water undersaturated

Bulk-rock composition

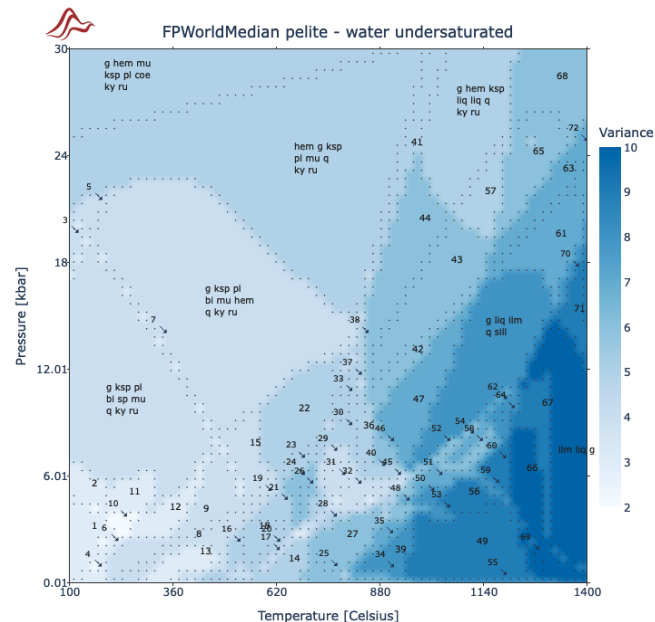
mol%

Drag and drop or select bulk-rock file

FPWorldMedian pelite - water undersatur^{ed}

oxide	fraction
SiO ₂	70.999
Al ₂ O ₃	17.005

Compute diagram



- Note that uniform grid and 6 subdivisions (4225 points) is enough for a density diagram for LaMEM
- It is not necessary to refine along reaction lines as only density is used

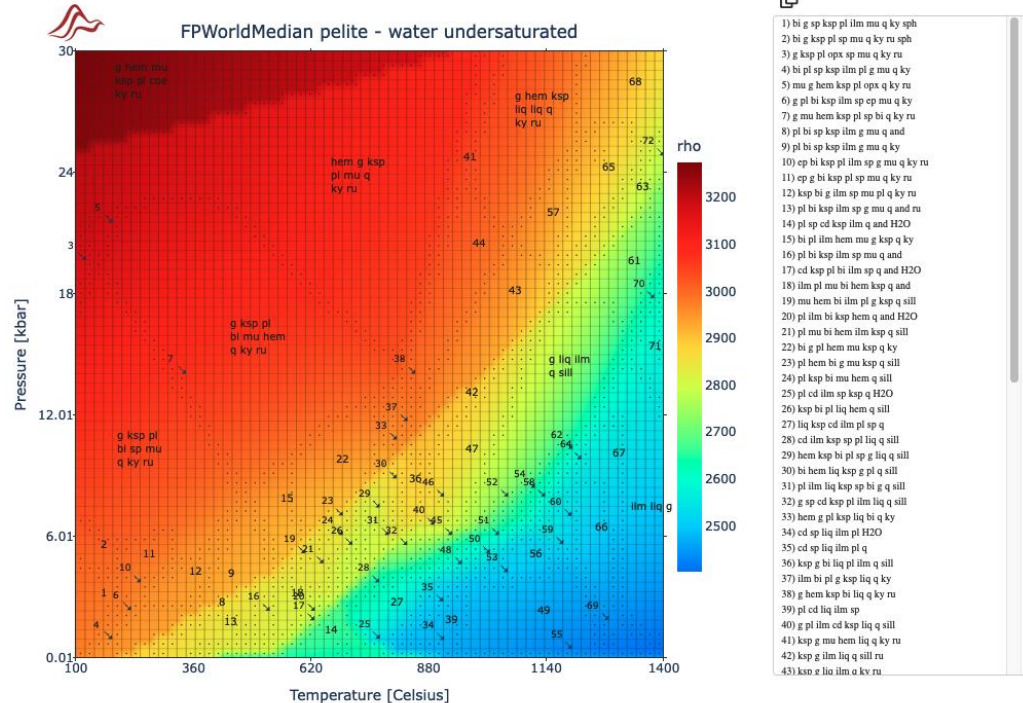
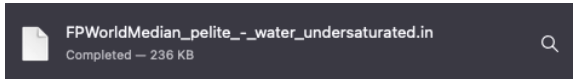
MAGMinApp: export to LaMEM

- Display system density and adjust visualization



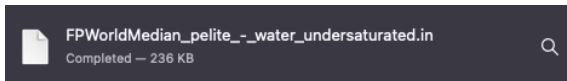
- Click “Export rho for LaMEM”

- This saves a “.in” file:



MAGEMinApp: LaMEM phase diagram file

- Input the “.in” file in a text editor



- The density diagram input format for LaMEM includes various information:

line 1	= number of columns starting line 56
lines 2 to 49	= comments
lines 50 to 55	= grid information
lines 56 to end	= data

- Phase diagram can be called in the LaMEM Julia script as follow.

```
1 mantle = Phase(      Name      = "Mantle",      # let's call phase 0 mantle
2                        ID          = 1,            # not that ID here points to
3                        alpha       = 3e-5,
4                        k           = 3,            # conductivity
5                        Cp          = 1000,         # heat capacity
6                        rho         = 3300,         # set mantle density
7                        disl_prof   = "Dry_Olivine_disl_creep-Hirth_Kohlstedt_2003",
8                        diff_prof   = "Dry_Olivine_diff_creep-Hirth_Kohlstedt_2003",
9                        G            = 5e10,
10                       rho_ph      = "../Peridotite" ); # set phase diagram
```

Note that the .in extension is absent!

MAGEMinApp other phase diagrams modes

MAGEMinApp: P-T phase diagram and trace-elements

- Select the following setup (nearly default)

Phase diagrams | **PTX path** | Isentropic path | General information

Setup | **Diagram** | Trace-elements

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18) ▼

Solution phase selection: ☐

Diagram type: P-T diagram ▼

H₂O-saturated at solidus: false ▼

☐ Limit Ca-opx: 0.5 ▼

TE predictive model: true ▼

Kd's database: O. Laurent (2012) ▼

Zr saturation: Crisp and Berry (2022) ▼

Pressure [kbar]: min 0.01 ▼ max 20.01 ▼

Temperature [°C]: min 800 ▼ max 1400 ▼

Initial grid subdivision: 3 ▼ 8 × 8 grid

Refinement type: Phases only ▼

Refinement levels: 2 ▼

Buffer: no buffer ▼

Solver: Hybrid ▼

Verbose: none ▼

Specific Cp: G0 ▼

Bulk-rock composition

mol% ▼

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous ▼

oxide	fraction
SiO ₂	38.494
Al ₂ O ₃	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K ₂ O	0.01
Na ₂ O	0.25
TiO ₂	0.1
O	0.096
Cr ₂ O ₃	0.109
H ₂ O	0

ig

Trace-element composition

Drag and drop or select trace-element file

Pyrolite (Primitive Mantle) ▼

Trace-element composition

Drag and drop or select trace-element file

Pyrolite (Primitive Mantle) ▼

elements	μg/g
Rb	0.6
Ba	6.6

- Select Pyrolite for trace-element composition

General parameters

Title: KLB1 Peridotite - Anhydrous

Update Reset

Compute phase diagram

Save/Load Diagram

filename Save state Load state

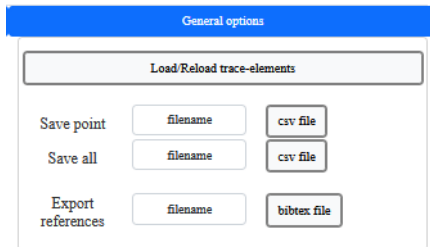
- Click compute phase diagram

MAGEMinApp: trace elements

- Click on “Trace-elements” tab



- Then on the right panel, click “Load/Reload trace-elements”



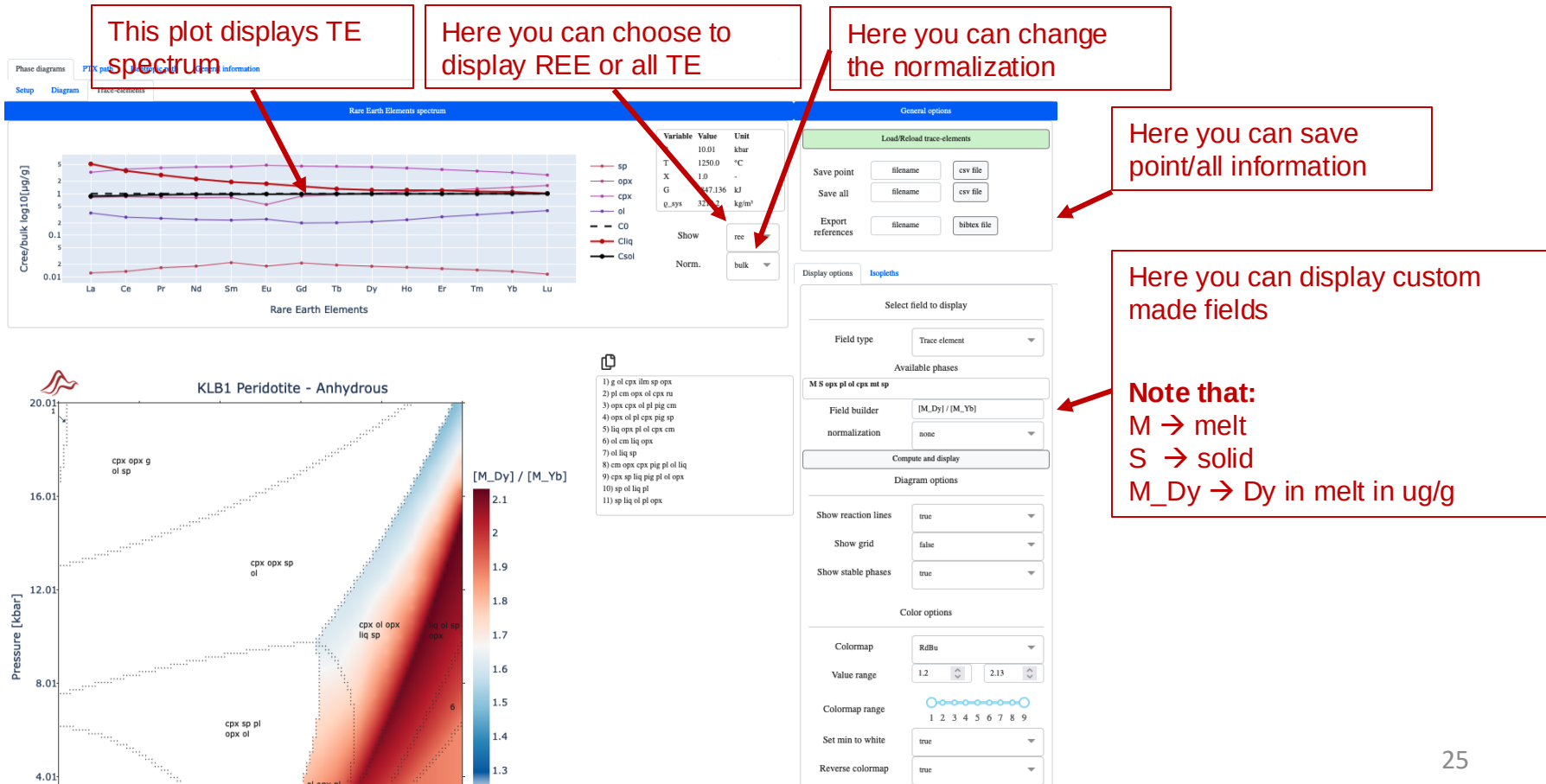
The screenshot shows the 'Display options' panel. It has a blue header with the text 'Display options' and a sub-header 'Isopleths'. Below the header is a section titled 'Select field to display'. This section contains a 'Field type' dropdown menu set to 'Trace element', an 'Available phases' section with a text input 'M S opx pl ol cpx mt sp', a 'Field builder' input '[M_Dy] / [M_Yb]', and a 'normalization' dropdown menu set to 'none'. Below these is a button labeled 'Compute and display' which is highlighted with a red rectangle. Below the button is a section titled 'Diagram options' containing three dropdown menus: 'Show reaction lines' set to 'true', 'Show grid' set to 'false', and 'Show stable phases' set to 'true'. Below these is a section titled 'Color options' containing a 'Colormap' dropdown menu set to 'RdBu', a 'Value range' section with two input fields '1.2' and '2.13', a 'Colormap range' section with a color bar and numbers 1 through 9, a 'Set min to white' dropdown menu set to 'true', a 'Reverse colormap' dropdown menu set to 'true', and a 'Smooth colormap' dropdown menu set to 'fast'.

- Then in display option, Field type → “Trace element” then click “Compute and display”
- Update the “Color options” as shown here
- Then click on anywhere in supra-solidus fields

MAGEMinApp: trace elements

Implemented with the help of J-F Moyen (LMV) and H. Dominguez (Uni - Bern)

- Which yields:



MAGEMinApp: custom bulk rock composition

- In Tab “Phase diagrams”, “Setup” select “Metabasite” then drag and drop \Resources\Mainz2024\bulkrock_composition_input_files\bulk-rock_ref.dat in the middle panel “Bulk-rock composition”

Phase diagram parameters

Thermodynamic database

Metabasite (Green et al., 2016)

Solution phase selection

Diagram type

P-T diagram

H₂O-saturated at solidus

false

clinopyroxene

Omphacite

☒

Augite

TE predictive model

false

min

max

Pressure [kbar]

0.01

16.01

min

max

Temperature [°C]

500

1200

Initial grid subdivision

3

8 × 8 grid

Refinement type

Phases only

Refinement levels

4

Buffer

no buffer

Solver

Hybrid

Verbose

none

Specific Cp

G0

Bulk-rock composition

mol%

Drag and drop or select bulk-rock file

Simon

oxide	fraction
SiO2	61.5415
Al2O3	8.5342
CaO	9.2213
MgO	5.0289
FeO	7.2396
K2O	0.2091
Na2O	3.4157
TiO2	1.2448
O	0.7269
H2O	2.8381

Metabasite (Green et al., 2016)

- In the bulk-rock dropdown menu, select “Simon” bulk

- Change the water content to 50 to oversaturate

MAGEMinApp: diagram with saturated water at solidus

Implemented with the help of J. Forshaw (Uni - Bern)

- **Select “H2O-saturated at solidus” = true**
This activates automatic saturation of water at solidus
- **Select Augite for the suprasolidus conditions**
- **Choose to activate or not TE modeling**
- **Choose the range of PT conditions you want**
- **Select a level of refinement according to your computer performances**
- **Then compute the diagram...**

Phase diagram parameters

Thermodynamic database: Metabasite (Green et al., 2016)

Solution phase selection: ☐

Diagram type: P-T diagram

H₂O-saturated at solidus: true

clinopyroxene: Omphacite ☒ Augite

TE predictive model: false

Pressure [kbar]: min 0.01 max 16.01

Temperature [°C]: min 500 max 1200

Initial grid subdivision: 3 8 × 8 grid

Refinement type: Phases only

Refinement levels: 4

Buffer: no buffer

Solver: Hybrid

Verbose: none

Specific Cp: G0

General parameters

Title: Simon

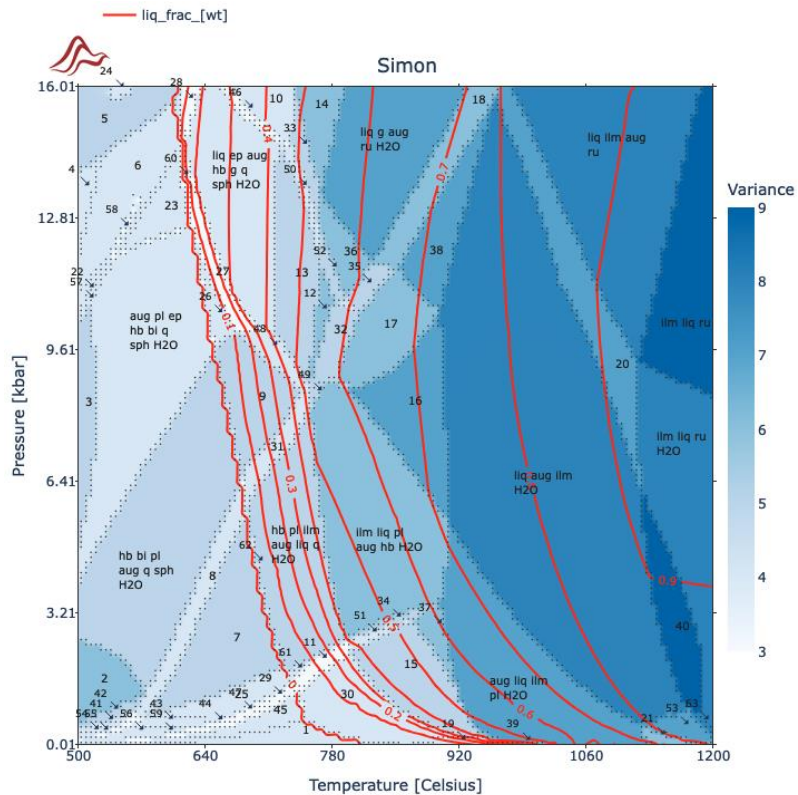
Update Reset

filename Save state Load state

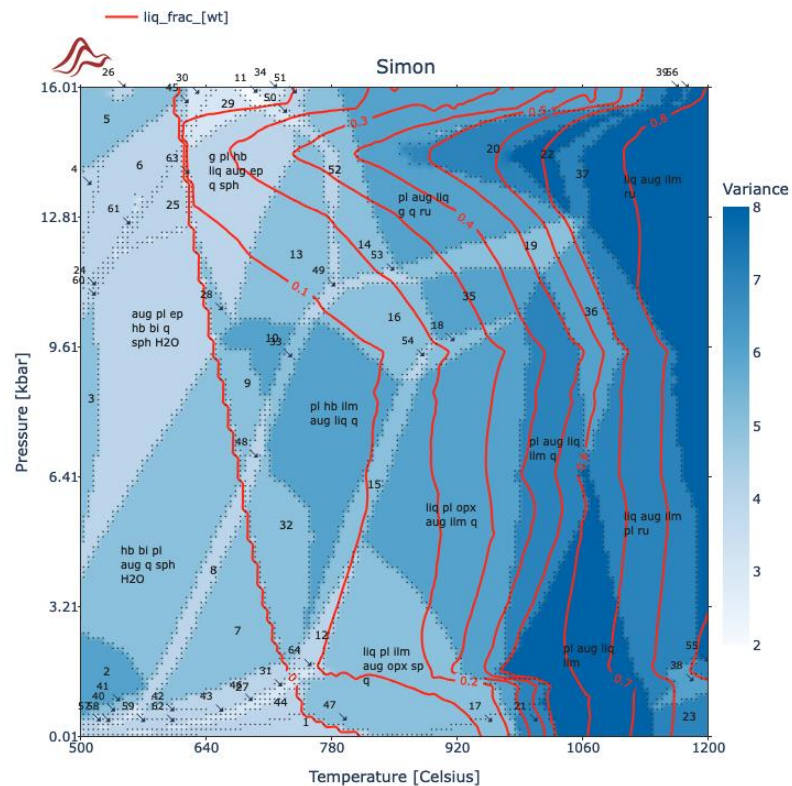
Compute phase diagram

State/CSV directory: /home/sepb42/MAGEMinApp/1/v0.4.1

MAGMinApp: diagram with saturated water at solidus



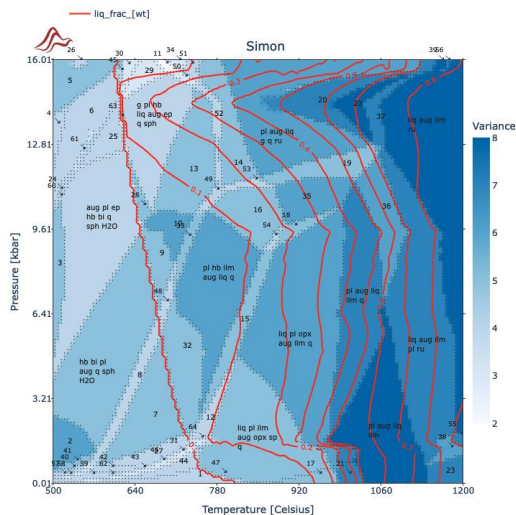
“H2O-saturated at solidus” = false



“H2O-saturated at solidus” = true

MAGMinApp: diagram with saturated water at solidus

- You can check the water-saturation at solidus by plotting system H₂O-activity (a_{H_2O})



"H₂O-saturated at solidus" = true



Informations Display options Isopleths Classifications

Select field to display

Field

Diagram options

Show reaction lines

Show stable phases

Show grid

Color options

Colormap

min max

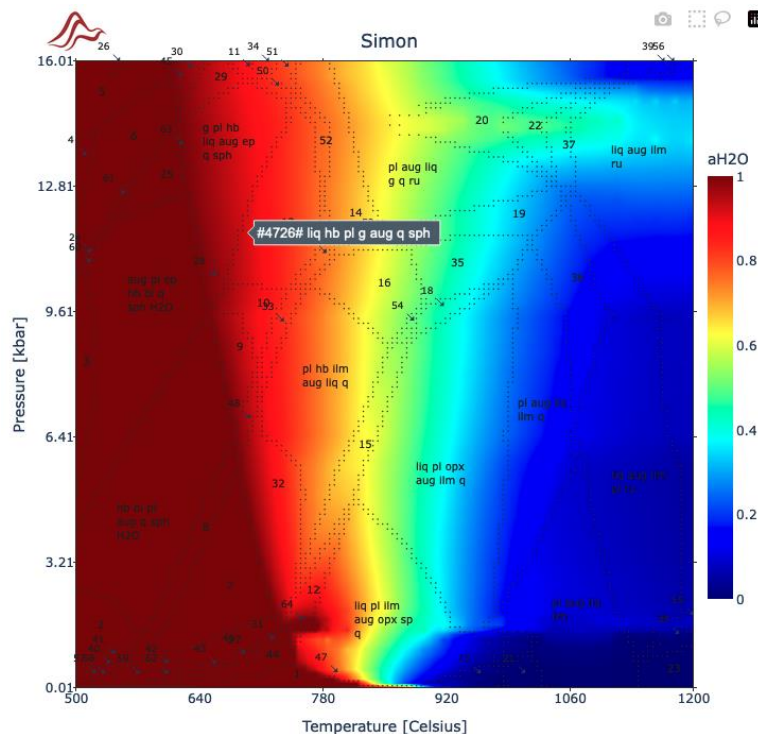
Value range

Colormap range

Set min to white

Reverse colormap

Smooth colormap



"H₂O-saturated at solidus" = true
+ few more refinement levels

MAGEMinApp: deactivating a solution phase model

Phase diagram parameters

Thermodynamic database: Metabasite (Green et al., 2016)

Solution phase selection

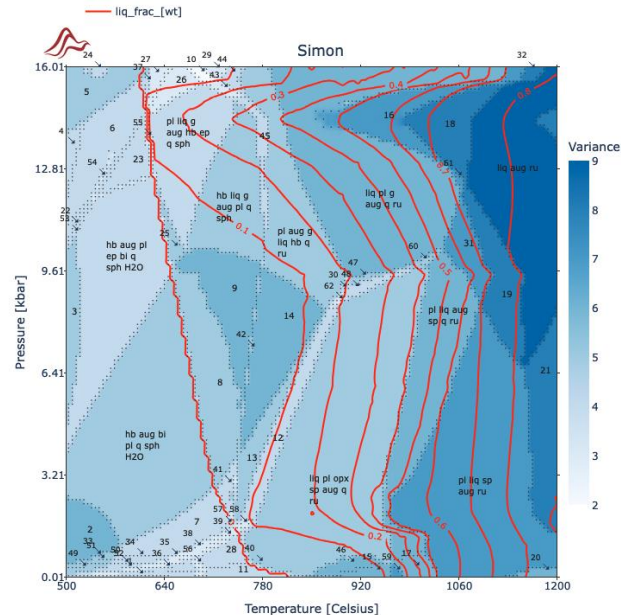
☐ ☒ sp ☒ opx ☒ fsp ☒ liq ☒ mu ☐ ilmm ☐ ilm ☒ ol ☒ hb ☒ ep ☒ g ☒ chl ☒ bi ☐ dio ☒ aug ☒ abc ☐ spn

Diagram type: P-T diagram

H₂O-saturated at solidus: true

clinopyroxene Omphacite ☒ Augite

- Go back to setup then click on “Solution phase selection” to unfold the list of solution phase model belonging to the “Metabasite” database
- Unselect ilm, ilmm, dio and spn
- Re-compute the diagram



“H2O-saturated at solidus” = true

MAGMinApp: T-X phase diagrams

- Open \\Resources\\Mainz2024\\bulkrock_composition_input_files\\bulk-rock_ref.dat in a text editor
- Copy the “Simon” composition and change the water content to 50.0 (do not add “;” at the end)
this provides the second bulk rock composition that will be used in the T-X diagram

```
# HEADER
title; comments; db; sysUnit; oxide; frac; frac2
# BULK-ROCK COMPOSITION
Test 2;Moo et al., 2000;ig;mol:[SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, Cr2O3, H2O];
[48.97, 11.76, 13.87, 4.21, 8.97, 1.66, 10.66, 1.36, 1.66, 0.01, 5.0];
K167;Moo et al., 2000;ig;mol:[SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, Cr2O3, H2O];
[50.3162, 8.0778, 11.6616, 11.3262, 9.1893, 0.4312, 2.4051, 2.271, 4.312, 0.0096, 0.0];
Test 3;Coin & Kwak, 1984;mb;wt:[SiO2, Al2O3, CaO, MgO, Fe2O3, K2O,Na2O, TiO2, FeO, H2O];
[55.12, 12.76, 4.32, 5.21, 2.45, 1.66, 10.66, 1.36, 1.66, 2.0];
Simon;Shorn Simon, 2023;mb;mol:[SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, H2O];
[61.8,8.57,9.26,5.05,7.27,0.21,3.43,1.25,0.73,2.85];
[61.8,8.57,9.26,5.05,7.27,0.21,3.43,1.25,0.73,50]
```

- In “Setup” tab load the modified bulk-rock_ref.dat
- Change “Diagram type” to “T-X diagram”
This displays a new table in the central “Bulk-rock composition” panel
- Select Simon for both  Notice that the right table automatically loads the second “Simon” bulk-rock composition

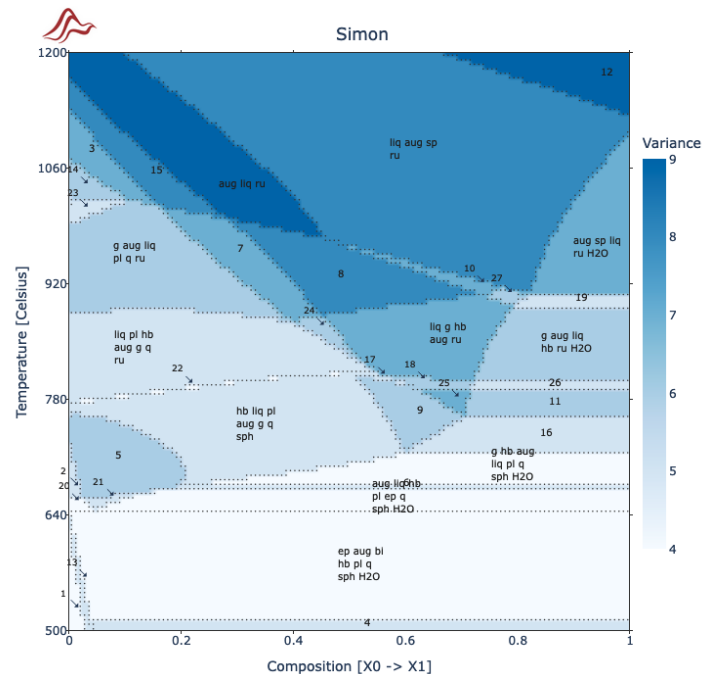
Bulk-rock composition			
mol% ▾	Drag and drop or select bulk-rock file		
Simon ▾		Simon ▾	
oxide	fraction	oxide	fraction
SiO2	61.5415	SiO2	41.8784
Al2O3	8.5342	Al2O3	5.8074
CaO	9.2213	CaO	6.275
MgO	5.0289	MgO	3.4221
FeO	7.2396	FeO	4.9265
K2O	0.2091	K2O	0.1423
Na2O	3.4157	Na2O	2.3243
TiO2	1.2448	TiO2	0.8471
O	0.7269	O	0.4947
H2O	2.8381	H2O	33.8822
Metabasite (Green et al., 2016)			

MAGEMinApp: T-X phase diagrams

- Change the Phase-Diagram setup according to what you want
- Then compute the phase diagram

Phase diagram parameters

Thermodynamic database	Metasite (Green et al., 2016) ▼	
Solution phase selection	☐	
Diagram type	T-X diagram ▼	
clinopyroxene	Omphacite ☒	Augite
TE predictive model	false ▼	
Temperature [°C]	min 500 ▼	max 1200 ▼
Fixed pressure	10.01 ▼	
Initial grid subdivision	3 ▼	8 x 8 grid
Refinement type	Phases only ▼	
Refinement levels	4 ▼	
Buffer	no buffer ▼	
Solver	Hybrid ▼	
Verbose	none ▼	
Specific Cp	G0 ▼	



Example of phase diagram result using 4 refinements levels

id=21eb31cc-0b24-4f38-bcd0-52daabb17c39; MAGEMinApp GUI v=0.6.2; using 12/12 threads

Set of parameters example

MAGEMinApp: T-X phase diagrams with buffer

- You can also change the buffer value in the RT log scale
- For instance, select QFM buffer

Phase diagram parameters

Thermodynamic database: Metabasite (Green et al., 2016)

Solution phase selection: ☐

Diagram type: T-X diagram

clinopyroxene: Omphacite ☒ Augite

TE predictive model: false

Temperature [°C]: min 500 max 1200

Fixed pressure: 10.01

Initial grid subdivision: 3 8 x 8 grid

Refinement type: Phases only

Refinement levels: 4

Buffer: no buffer

Solver: Hybrid

Verbose: none

Specific Cp: G0

- Then select SM89 bulk-rock for both tables
- Oversaturate in oxygen (O)
- change the buffer multiplier values

Bulk-rock composition

mol% Drag and drop or select bulk-rock file

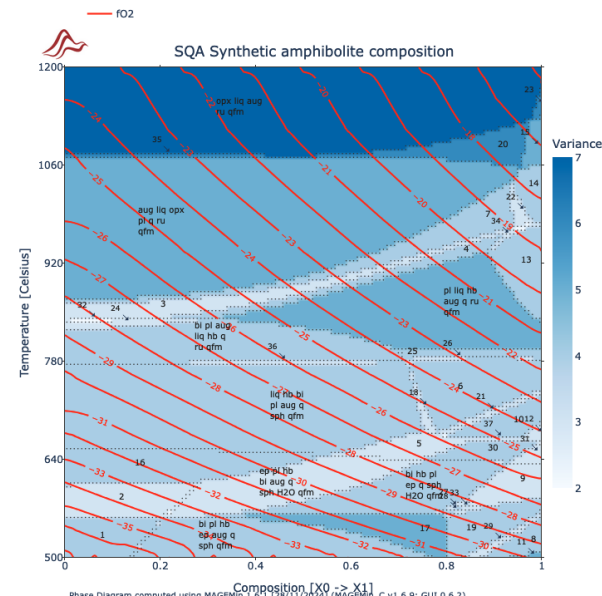
SQA Synthetic amphibolite composition

oxide	fraction	oxide	fraction
SiO2	60.05	SiO2	60.05
Al2O3	6.62	Al2O3	6.62
CaO	8.31	CaO	8.31
MgO	9.93	MgO	9.93
FeO	6.57	FeO	6.57
K2O	0.44	K2O	0.44
Na2O	1.83	Na2O	1.83
TiO2	1.27	TiO2	1.27
0	3	0	3
H2O	4.64	H2O	4.64

Buffer multiplier: -4 4

Metabasite (Green et al., 2016)

- Compute and contour the field log₁₀(fO₂) (Oxygen fugacity)



id=21eb31ce-0b24-4f38-bed0-52daabb17c39; MAGEMinApp GUI v=0.6.2; using 12/12 threads

Set of parameters example

MAGEMinApp: PT-X phase diagrams

- PT-X diagrams allow you to define a PT path on the vertical axis and vary the composition in the horizontal axis
- Let's compute a Barrovian PT path phase diagram for metapelite while varying the water content
- First select the "Metapelite" database, then change Diagram type to "PT-X diagram"

Phase diagram parameters

Thermodynamic database

Metapelite (White et al., 2014)

Solution phase selection

☐

Diagram type

PT-X diagram

TE predictive model

false

Pressure-Temperature path
(using pChip interpolation)

	P [kbar]	T [°C]
×	3	200
×	6	400
×	8	600
×	9	800
×	8	850
×	6	700
×	3	400

Add new point

Initial grid subdivision

3

8 x 8 grid

Refinement type

Phases only

Refinement levels

4

Buffer

no buffer

Solver

Hybrid

Verbose

none

Specific Cp

G0

Bulk-rock composition

mol%

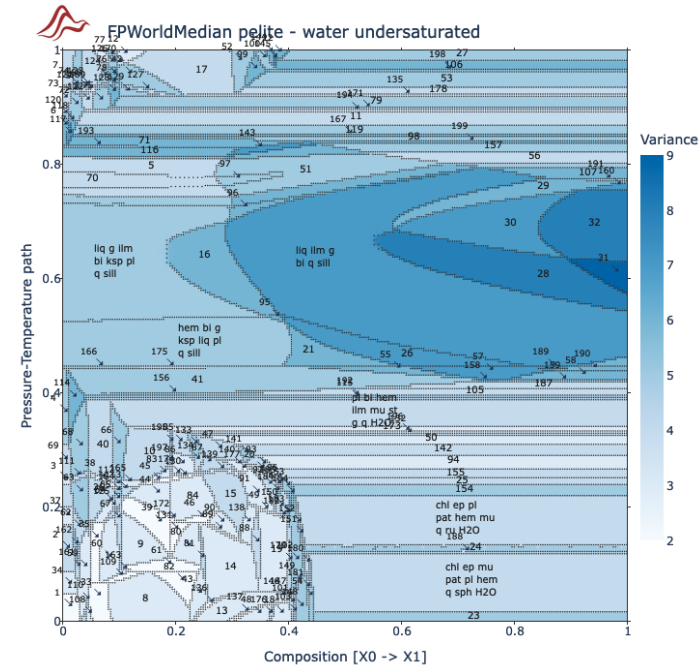
Drag and drop or select bulk-rock file

FPWorldMedian pelite - water undersatur

FPWorldMedian pelite - water oversatur

oxide	fraction	oxide	fraction
SiO2	70.999	SiO2	70.999
Al2O3	12.805	Al2O3	12.805
CaO	0.771	CaO	0.771
MgO	3.978	MgO	3.978
FeO	6.342	FeO	6.342
K2O	2.7895	K2O	2.7895
Na2O	1.481	Na2O	1.481
TiO2	0.758	TiO2	0.758
O	0.72933	O	0.72933
MnO	0.075	MnO	0.075
H2O	5	H2O	30

Metapelite (White et al., 2014)



Note that if the field's list is cropped in the app, you can access the complete list of stable mineral assemblage in the "svg" when exporting the figure